RL-Based Adaptive Task-Offloading in Mobile-Edge Computing for IoT Networks

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Abstract

The Internet of Things (IoT) has been increasingly used in our daily lives and in industry. However, due to limitations in computing and power capabilities, IoT devices that need to send their respective tasks to the cloud service stations, which are usually located at far distances. Having to transmit data over long distances presents challenges for services that require low latency, such as industrial control in factories, as well as autonomous driving assisted by artificial intelligence. To solve this issue, mobile edge computing (MEC) is deployed at the network's edge to reduce transmission time. This study proposes a new offloading scheme for MEC-assisted ultra-dense cellular networks using reinforcement learning (RL) techniques. The RL algorithm learns from the historical data of the network and adapts the offloading decisions to optimize the overall performance of the network. Non-orthogonal multiple access is also adopted to improve resource utilization among IoT devices. The simulation results demonstrate that the proposed scheme outperforms other state-of-the-art offloading algorithms in terms of energy efficiency, network throughput, and user satisfaction.

I. Introduction

Recently, Internet of Things (IoT)-enabled 6G networks and computing paradigms have gained a lot of attention both in industry and academia, where Mobile Edge Computing (MEC) has emerged with the goal of providing cloud power at the network edge, addressing key issues such as connectivity, latency, and bandwidth that Cloudlet Computing (CC) cannot handle on its own [1], [2]. An example of addressing this issue, MEC brings computing and storage resources closer to the edge of the mobile network, allowing highly demanding IoT-based applications to run on devices while still maintaining delay constraints [3], [4]. Another advantage of MEC confirming its resource-efficient policy is that its resources could also be exploited by other nearby entities, such as operators and third parties for performing specific purposes [5]. By incorporating intelligence into edge devices, also known as edge nodes, edge computing acts as a distributed computing concept that enables real-time data processing and analysis near to the point of data aggregation. Data does not have to be directly uploaded to the cloud or a centralized data processing system with edge computing [6]. By adopting edge computing, most businesses use centralized storage, often in a public cloud or private cloud environment, to store, manage, and analyze their data [7]. Unfortunately, many real-world applications no longer have the needs that can be satisfied by traditional infrastructure and cloud computing. Recently, the use of MEC and energy harvesting (EH) has become increasingly popular in the IoT domain. MEC provides vital functions such as computations processing, and storage capabilities through edge servers, which are usually placed as near as possible to mobile user devices, while EH provides a means of prolonging to extend the life of IoT devices by continuously providing energy [8], [9]. However, effective offload schemes are needed to optimize the performance of MEC-based IoT devices. Edge computing eliminates the need to transport data to the cloud or an on-premises data center for processing and analysis, because data are processed near the data collection source [10]. The congestion in the networks and servers will be reduced by this strategy. Edge computing is also very useful in industrial IoT (IIoT), due to its faster response time and capacity for real-time data processing [11]. The importance of investigating task scheduling and offloading arises from several aspects, such as task arrivals possessing a heterogeneous nature of dynamics that is experienced by the MEC's nodes, and the distributed nature of the overall MEC system. Hence, the accumulated workloads of the MEC's edge nodes are more likely to be prone and pass through unbalanced phases, leading to a high response time, and a high resource cost [12]. Regarding the contribution that nonorthogonal multiple access (NOMA) offers to the process of task offloading in MEC, the non-orthogonal property of NOMA enables it to maintain successful implementation of the MEC and the offloading process of the computation's tasks to MEC servers by optimizing the adopted resource allocation approach. This is very important because NOMA achieves efficient resource usage, which, in turn, is vital for achieving efficient task

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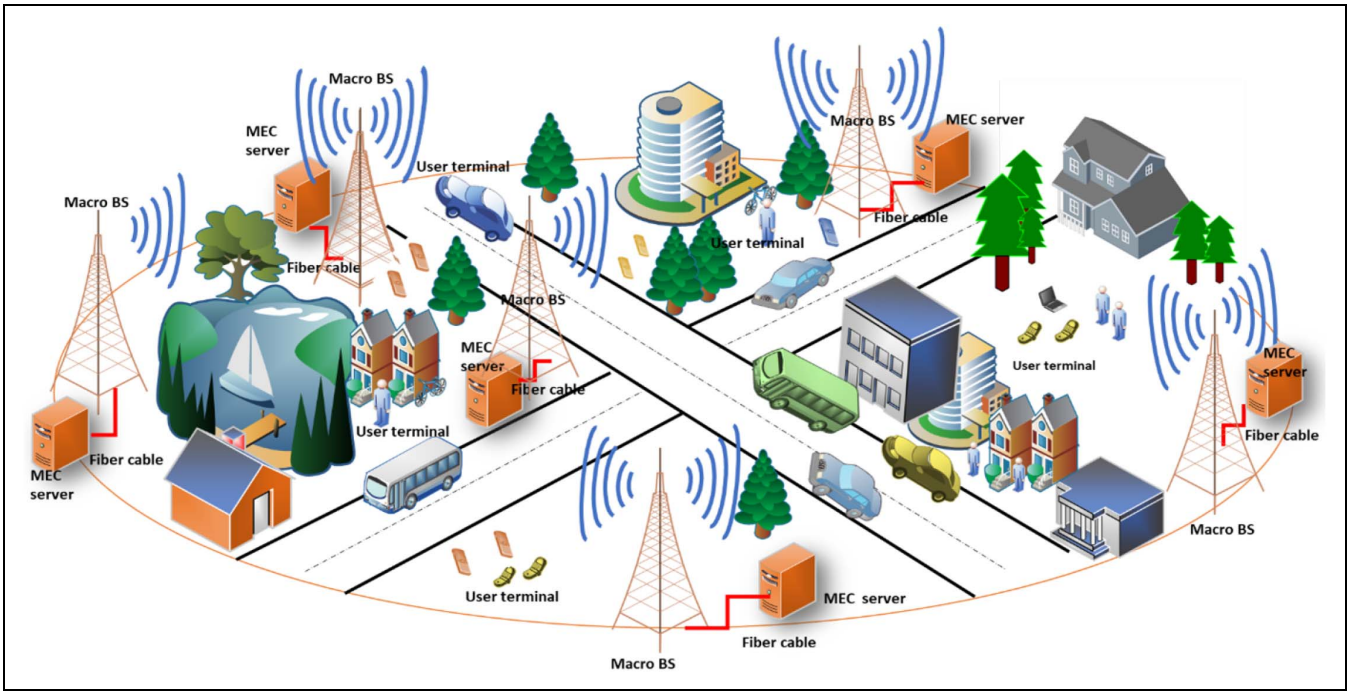


FIG. 1. An illustration of the considered system model of MEC based IoT enabled ultra dense network.

offloading. According to the idea of superposition coding, different power levels and coding schemes are assigned to different users [13]. In the MEC context, MEC servers suffer from computational resource limitations. Therefore, adopting NOMA first ensures that multiple mobile users can offload their computation tasks simultaneously to the MEC server without enduring severe competition over the available resources. Secondly, NOMA contributes to balancing the allocation of computing resources among mobile users, which is necessary, especially in dense user scenarios [14], [15].

Latency in any wireless communication system is considerably affected by factors such as the achievable data rate and the overall reliability of the system over the communication link. Hence, NOMA could be used to optimize the data rate and reliability of each mobile user. As low latency is vital for practical applications in MEC, it is possible to enhance the achievable data rate, which, in turn, reduces the latency for task offloading through the optimization of the allocated power and code rate for each NOMA user. Among the techniques considered, reinforcement learning (RL)-based approaches appear to be a popular method for tackling the challenges of MEC, including the management of limited resources and the optimization of offloading decisions; hence, it inspires the work of this paper to adopt the RL approach. Although various mechanisms for optimizing offloading policy in MEC-based IoT devices appeared in the open literature, there are still some gaps and areas that future research could investigate. In light of this, the contributions of this work are as follows:

- A new optimized adaptive offloading strategy is proposed that allows mobile devices to learn optimal offloading decisions in various scenarios is proposed. A multi-agent deep Q-network is considered to learn and then reach the optimal offloading decision in a distributed manner.
- The proposed approach outperforms traditional offloading methods in terms of latency and energy consumption. The proposed scheme selects the edge device and the offload rate based on each server workload.
- A DRL-based offloading scheme is also proposed to enhance the speed of the learning process.
- Proposing an RL-based mobile offloading scheme for MEC that uses safe RL to avoid adopting an unstable offloading procedure in cases where it fails to achieve the predetermined goals within the system constraints such as the task's computational latency. The proposed strategy allows each mobile device is to select the best edge device, offloading rate and transmit power to optimize its own utility without the need for knowledge of the model related to task generation, MEC, or even the interference model.

II. The Proposed System Model

The proposed system model consists of M base stations (BSs) indexed as $m_1, m_2, m_3, \ldots, m_M$. The modeled system environment is based on LTE standards that are set up to meet the expected demands of future generations, as shown in Fig. 1, with NOMA chosen as the air interface. We adopted the concept of a nearest–farthest user-pairing scheme in which the strong and weak channel users simultaneously share the same resource block. The power is allocated by using a proportional power ratio to the reverse of each user's channel power so that $(P_{weak} > P_{strong})$ is always maintained. This is important to guarantee the successful application of successive interference cancellation (SIC). As a QoS metric, a minimum rate threshold is adopted as a QoS constraint for each user to ensure that each user achieves a certain level of target data rate. It is assumed that in each BS, m, there is an edge server; therefore, the delay between the two could be counted as zero. These servers are indexed as $s_1, s_2, s_3, \ldots, s_M$. Each edge server is characterized by a computational capacity of a certain limit. The total

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number of users is set to K, all of which generate a total of T computing tasks.

The calculations are modeled so that, for the i-th user that has a computation task τ_i, the input data has the size of D_i. On the other hand, the total CPU cycles that are required for processing each single bit of data are denoted by S_i. The offloading decision of each user is modeled as $A = \{a_{ij}\}_{K*(M+1)}$, and two computational strategies are considered in this paper. The case where $a_{(i,j)} = 1$ refers to the user task is being offloaded to the BS. On the other hand, the case where $a_{(i,0)} = 1$ indicates the task is executed locally. Task execution through offloading is applied when the user sends the computing task for processing to the intended BS. After that, the data computation process is applied by the edge server of that BS. Finally, the BS returns the results obtained to the same user. The duration of time consumed at this stage represents a very small percentage of the transmission time period. Since there is no data transmission at the local execution of the computing tasks, the energy consumed will only be represented by the computing energy, and its value depends on the chip architecture. On the other hand, the energy consumed in the case of offload execution takes into account the energy consumed during the computation phase and the transmission phase. The objectives considered to be maintained by the proposed system model are to minimize both the energy consumed and the time delay during the processing of each task. Achieving such objectives will facilitate maintaining a high degree of Quality of Service (QoS) under the constraint of limited energy in the BS. The first problem has the objective of minimizing the total latency and the overall energy consumption by optimizing the offloading decision for each task.

In the considered optimization problem, the primary optimization objective is the joint minimization of energy consumption and task latency. In particular, latency is treated as an optimization component in the weighted objective function and also treated as a QoS constraint and it is considered to be upper-bounded by the maximum tolerable delay. The proposed problem is formulated as a mixed-integer optimization problem using integer-valued weights of the time elapsed during task processing and the energy consumption weight (assuming that they are termed as w^a and w^b), respectively. The integer values of those two weights sum up to an integer, constant value termed as ϕ. For example, if $w^a = w^b$, the offloading within the considered communication scenario assumes that both latency and energy consumption are equally important. Moreover, if $w^a > w^b$, the scenario considered is assumed to be more sensitive to latency. Otherwise, if $w^a < w^b$, the scenario is assumed to be more sensitive to energy consumption. It must be noted that the formulated problem gives insight into the trade-offs between several aspects, i.e., energy consumption, latency, each user capacity, the edge server capacity, and the communication capabilities. By examining and analyzing the results, it is possible to understand how these factors play a vital role in affecting the overall performance of the system and how they contribute to making informed decisions and consequently optimizing the system performance. Solving the proposed optimization problem and finding the optimal solution is possible by applying the Mixed Integer Linear Programming (MILP) technique. This is because MILP permits formulating and solving the types of optimization problems that have both continuous and discrete variables, which makes it a good match for the MEC decision-offloading process.

For taking advantage of the MILP approach, several aspects of the optimization problem need to be predefined, such as decision variables, the objective function, and constraints. In this paper, the problem is solved numerically using the MILP solver toolbox in MATLAB, in particular, the intlinprog function is adopted to find the optimal solution. To minimize the overall energy consumed during task execution. The second proposed optimization problem has the objective of minimizing the energy consumption to execute tasks while maintaining the latency constraint. This is important due to the scarcity of power resources and the need to establish an energy-efficient system. The objective function minimizes the total energy consumption for executing all tasks, and the constraints guarantee that the latency experienced by each mobile user to process each computing task is within the QoS limits.

III. Implementation of Q-Learning for Task Offloading in Edge Computing

This section presents a task-offloading problem that is based on the tabular Q-learning algorithm. The problem setup includes a set of mobile users and a set of BSs, each of which is assumed to be equipped with an edge server. Each user is assumed to have a set of tasks to be processed. These tasks could be executed locally on the user device or offloaded to an edge server. The goal is to find the best offloading decision for each task and each user to minimize the total execution time of the tasks and hence minimize the latency. The Q-learning algorithm is adopted here to assist in learning and make optimal offloading decisions over time. Therefore, the approach in this section is not formulated as a learning problem but rather as an optimization problem. The objective here is to obtain the optimal policy that maximizes the total reward obtained by the agent to make the optimal offloading decision, taking into account the resulting consequences and the acquired rewards.

A. The Proposed RL-Based Approach

In each time slot, the user's terminal sends a computing task, and those tasks follow a Poisson distribution. Depending on the capacity of the servers and their status, whether being overloaded or not, the computing tasks are processed either by those servers, and this is what is known as local processing, or else the tasks are offloaded. The main goal behind the learning problem is to optimize the offloading decisions of tasks of each mobile user so that their computation time and communication time are minimized. Minimizing the computation time and communication time could directly contribute to maximizing the rewards acquired by those users. To be precise, the proposed decision-making process is applied through sequential steps. The goal of this algorithm is to find the edge server capacity that maximizes the total reward of all users. Firstly, it computes the valid off-loading decisions for each mobile user and each BS's edge server. This is done based on the computation and communication times of the tasks and the edge server capacity. Then the decision-making algorithm picks the next task allocation for each user based on the valid off-loading decisions. Finally, it computes the

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reward for each user based on the total time required to complete the task.

B. The Proposed Modified Tabular Q-Learning Algorithm

In this section, the tabular Q-learning algorithm is exploited, which works on updating the Q-value table by mapping the states and actions to their respective rewards. It stores the values in a table where the rows represent the states, and the columns represent the actions. The Q-learning algorithm stores the predicted rewards in its Q-table, which is iteratively updated. These rewards help each agent perform a certain action in a certain state. In order to continuously optimize its own cumulative rewards, each agent is responsible for monitoring its environment and the receipt of rewards. In conventional form, the formula that Q-learning uses for updating is given by

$$Q(\phi,\vartheta) \leftarrow Q(\phi,\vartheta) + \delta\Big[r + \beta \max_{a} Q(\phi',\vartheta) - Q(\phi,\vartheta)\Big] \tag{1}$$

where $Q(\phi,\vartheta)$ is the value of Q at the ϕ-th state and the ϑ-th action, ϕ is the current state, ϑ is the action taken by the agent, r is the reward acquired by the agent after taking the ϑ-th action in the ϕ-th state, β is the discount factor reflecting the importance of future rewards and ϕ' is the next state reached after the agent takes the ϑ-th action.

It should be noted that Q-learning sometimes acts less efficiently in environments characterized by large state spaces. In this paper, a modification is proposed to produce a more advanced strategy in terms of exploitation and exploration. In other words, while our current implementation includes a tabular Q-learning approach, it must be noted that as the number of users and servers increases, the state-action space grows considerably. To cope with this issue, we highlight that we grouped states with similar workload and similar channel conditions together. In such a case, the state-action space would still be manageable. It is also assumed that agents adopt the concept of the ε-greedy scheme. Each agent randomly monitors the premises with an exploration probability of ε and performs the best possible action with a probability of $(1-\varepsilon)$. Hence, after applying the proposed modification to (1), the updated equation of the Q-learning approach will be given as

$$Q(\phi,\vartheta) \leftarrow Q(\phi,\vartheta) + \delta\Big[r + \beta((1-\epsilon) \times \max_{a} Q(\phi',\vartheta) + \epsilon\tau) - Q(\phi,\vartheta)\Big] \tag{2}$$

where τ represents a random exploration weight that is also affected by the difference between the current reward value and the reward value previously assigned to the same agent. In such a case, the proposed rewarding strategy will encourage the agent's exploration to be focused on a direction that almost guarantees benefit in favor of the Q-value and hence, the whole system. After that, each agent updates the Q-values for each state-action pair based on the proposed approach in (2), which has a type of recursive formula that relates the value of a state to the values of its closest states. After applying (2), the reward value r is also stored to be checked after each action in such a way that if the acquired reward is worth the action taken by the agent, then the agent would proceed in its action. However, if the newly obtained state is not much different from the previous one, the reward would be set at the same value to encourage the agent to take better action. This is very important in dense scenarios such as ultra-dense IoT networks considered in this paper. The proposed RL-based algorithm then performs the selected actions and receives a reward for each end node.

C. The Proposed Reward Calculations

In the proposed RL-based algorithm, the reward is simply calculated as the negative quantity of the total execution time (total latency) for each user's tasks. In this case, and to obtain a precise result, it will include both computation times and the communication times. This includes both local computation time and communication delay. For instance, if a user task requires 1 ms for computations and 2 ms for transmission, then the reward to be given is determined as -3. This ensures that minimizing task completion time directly maximizes the cumulative reward. Introducing latency as a negative reward allows the Q-learning agent to naturally learn to select offloading actions that reduce both delay and energy consumption simultaneously.

The proposed learning algorithm then updates the Q-table according to the received rewards and the expected rewards for the next state and action. The Q value table is updated using a learning rate, a discount factor, and the received and expected rewards. It is worth mentioning that the algorithm repeats this process for a fixed number of episodes. After applying the proposed modified tabular Q-learning algorithm within the proposed approach, the algorithm determines the offloading decision surface, taking into consideration the system model environment, including the dense nature of the network. This is to show the predicted total reward for each offloading task and each corresponding edge server. Then the proposed algorithm counts the average reward received by every end node in all episodes and in all Monte Carlo simulation iterations.

IV. Simulations and Results

The simulated scenario emulates a dense situation where around 10 to 100 nodes are active in the covered area, and each one is assumed to have up to 10 tasks. The total number of servers is chosen to be 5. The communication speed is set as 10Mb/s while the speed of processing is set to around 1Gb/s and the data size is 1Gb/s. The carrier frequency of this set-up is set to 5GHz. The non-uniform allocation of the MEC serves and the nodes is adopted randomly for a more practical scenario.

Fig. 2 shows the QoS performance for the optimized off-loading, the local offloading approach that presumes to perform all the offloading locally, the proposed RL-enabled least load offloading, and the proposed RL-enabled offloading scheme. Note that QoS is calculated by taking into consideration the energy consumption and latency for each scheme.

The reliability of the compared schemes is examined in Fig. 3. The proposed optimized approach and the proposed RL-based offloading approach both offer comparable levels of reliability but lower than that of the proposed RL-enabled least load offloading

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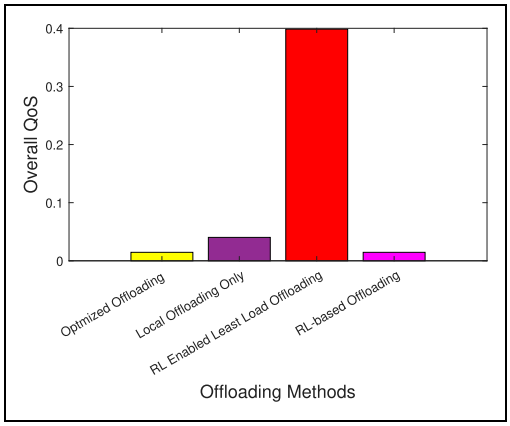


FIG. 2. Comparison of the proposed offloading schemes in terms of the level of QoS provided by each approach.

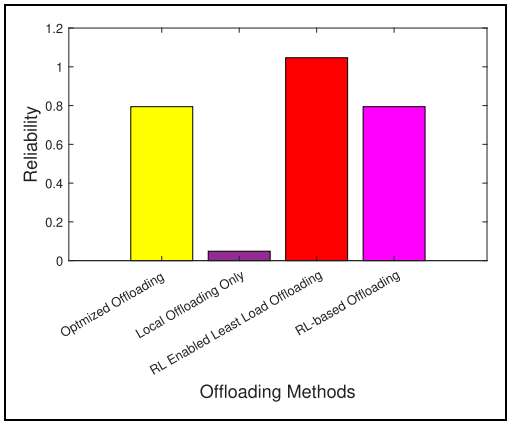


FIG. 3. Comparison among the proposed offloading schemes in terms of the level of reliability.

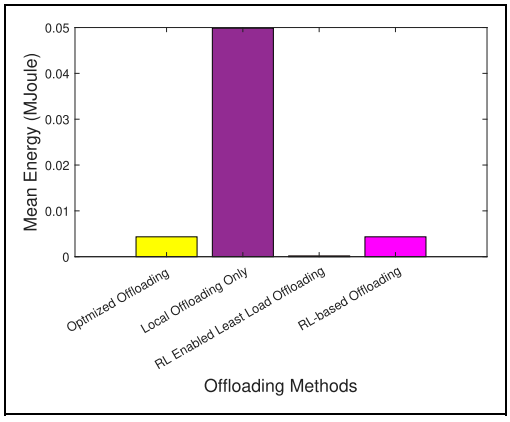


FIG. 4. Comparison among the proposed offloading schemes in terms of the average energy consumed by each approach.

scheme. All three of those schemes offer better reliability than the local execution approach. Fig. 3 also shows that the offloading scheme that depends on RL and chooses the server with the least load to perform the offloading offers the best QoS among all the schemes. This is mainly because of the attributes of the proposed RL that benefit from earlier learning of all servers' conditions before deciding on the offloading server. This leads to efficient energy usage and less latency spent in executing the offloading process.

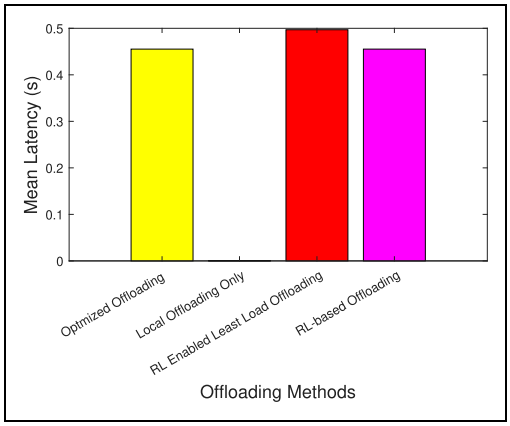


FIG. 5. Comparison of the proposed offloading schemes in terms of the average latency consumed by each approach.

The average energy consumption is shown in Fig. 4 for all the proposed schemes. This figure illustrates that the RL-enabled least offloading scheme consumes the least energy while offloading a task compared to the other schemes. This highlights the vital role of RL in choosing the offloading server, as well as the importance of selecting suitable reward and learning approaches.

Fig. 5 shows a comparison of the average latency of each of the proposed schemes against the local execution of the task offloading process. Examining the latency of each offloading approach is important, especially for 6G-enabled IoT network applications. Predictably, the local execution scheme clearly offers the least latency, since there is no delay or latency all related operations are performed locally. However, the proposed RL-enabled least load offloading strategy offers a slightly higher latency than the other two proposed schemes. This slight latency part could be the result of the learning and server-selection process, which could be reduced by selecting another learning or rewarding technique. All in all, the proposed RL enables the least load offloading scheme offers the best performance as compared to the proposed RL-based offloading and optimized offloading, with slightly higher latency. On the other hand, the proposed optimized offloading strategy shows promising behaviors in terms of energy consumption, latency, QoS and reliability but still falls behind the proposed RL-enabled load offloading scheme. Choosing the right offloading scheme might well depend upon the given application by taking into account energy consumption and latency as priorities.

V. Future Research Directions

Although the proposed offloading framework using optimization and RL demonstrates promising performance for MEC in wireless IoT networks, there are still several areas that need further research and development. In addition to improving system performance, these efforts seek to increase the robustness and practical application of MEC-based offloading solutions under actual constraints.

A. Sustainable and Energy-Efficient MEC Offloading

Typically, IoT devices have limited resources and the edge servers that themselves add to the total power consumption of the network. Therefore, green offload algorithms that reduce energy consumption while

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maintaining a satisfactory quality of services should be the focus of future research. Multi-objective optimization approaches that balance latency, throughput, and energy usage in offloading decisions could help achieve this.

B. Joint Optimization of Computation, Communication, and Caching

Similar to the majority of existing solutions, our current study focuses only in computing offloading. A more comprehensive strategy would optimize caching, communication, and computing simultaneously. By pre-fetching or storing popular material at the edge nodes, such a collaborative design could minimize unnecessary computations and data transmissions. Deep multi-agent RL frameworks that cooperatively manage dispersed resources or sophisticated graph-based RL techniques could be used to mimic this.

C. Federated Learning Integration for Privacy-Preserving Offloading

Integrating federated learning (FL) with offloading choices is a crucial future path. Centralized training in existing RL-based offloading systems can jeopardize user privacy, particularly in delicate IoT domains such as industrial monitoring or healthcare. Without exchanging raw data, FL enables edge devices to work together to train global models. It is worthwhile mentioning that integrating FL into the MEC offloading process will preserve learning effectiveness while protecting privacy.

D. Using Meta-RL to Maintain Quick-Adaptation

It is known that conventional RL agents may perform worse in dynamic wireless contexts due to user mobility or nonstationary network conditions. Using meta-RL, which enables agents to quickly adjust to novel tasks or surroundings with little data, is one possible approach. This could greatly increase the system's resilience in practical deployments by improving its responsiveness to erratic changes in network topology, load variations, or shifting user demands.

E. Deployment and Scalability in Diverse Environments

A wide range of devices, edge servers, and communication protocols are used in real-world MEC systems, which are extremely diverse. In order to solve scalability and interoperability concerns, future studies should create offloading models that are lightweight and flexible enough to function well on a variety of platforms. To handle such complexity, collaborative cloud-edge-device frameworks and hierarchical reinforcement learning could be crucial.

VI. Conclusion

This paper discussed how the use of RL-based adaptive task offloading in IoT networks and MEC-assisted ultradense networks can offer a promising solution for efficient and intelligent resource management in IoT networks. This paper also proved that leveraging the features of machine learning algorithms enables dynamic and adaptive task allocation, which contributes to improving the overall network performance, as well as maintaining reduced latency and enhancing the overall network scalability. In addition, exploiting NOMA boosts the successful implementation of MEC and task offloading through power optimization and efficient resource allocation. The results showed that there is a proportion between the compared entities. This is mainly because having more mobile users means more tasks to offload, and more energy to be consumed in the offloading process. Furthermore, a higher number of mobile users implies increased competition among users over the available resources.

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Biographies

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